BELEM CAPITAL

EPFL HACKATHON 2026
CHALLENGE TRACK

DeFi Risk Dependency Challenge

Turn a tangled web of on-chain dependencies into one number a portfolio manager can act on: stay, or get out.

WHO IS SETTING THIS CHALLENGE

Belem Capital is a Luxembourg-based on-chain asset manager and solution partner for professional and institutional investors, running market-neutral yield strategies across stablecoin, ETH, BTC and SOL mandates through a registered Luxembourg fund, segregated managed accounts and curated vaults.

Capital is deployed only on protocols that clear our pre-selection scorecard, and every position is monitored in real time. Security is not a side project here, it is the reason the track record exists. This challenge is one of the questions our own risk team works on.

~$400M Assets under management	**71 months** Track record since October 2020
0 · 0 · 0 Security incidents, counterparty defaults, liquidations	**39+ · 20+** Protocols and blockchains monitored

THE PROBLEM

One position, a dozen silent dependencies

On-chain positions are composable, which is another way of saying they inherit risk from everything underneath them. A single position can depend on several protocols, assets, oracles and infrastructure layers at once, and most of that chain is invisible from the position itself.

When one component is hit by a security incident, a market disruption or a technical failure, the hard part is not noticing the incident. It is working out what it does to your book, and how fast.

Same question for a vault routing capital into a perp venue, for a protocol concentrating a large share of a chain's TVL, or for an asset priced by a single feed.

Worked example: a Pendle PT position

A principal token is one line in a portfolio. Its dependency graph is five layers deep, and a failure at any layer prices through to the position.

- LAYER 1 — **Settlement chain**
- LAYER 2 — **Underlying yield source and its issuer**
- LAYER 3 — **Oracle pricing the asset** incident
- LAYER 4 — **PT market and its AMM curve**
- LAYER 5 — **Lending market holding it as collateral**

THE CHALLENGE

Build the system

Automatically identify the dependencies of a position, and estimate the potential economic impact on that position when one of them is affected.

You are free to choose how dependencies are discovered, how risk is modelled and how the final signal is produced. We care about the reasoning, not about a particular framework.

Answer the question a risk manager actually asks

Should we stay in the position, or is the expected loss high enough to justify exiting despite execution costs?

Take a book split across ten pockets of 10% each. One pocket is affected. Unwinding it costs 11% in slippage. Exiting is expensive and certain, staying is cheap and uncertain. A useful system makes that trade-off explicit instead of leaving it to instinct at three in the morning.

SCOPE AND FREEDOM

- Any ecosystem: Solana, EVM chains, or newer settlement layers such as Canton or Tempo
- Oracle dependency and feed-quality analysis
- Vaults that deploy into an external venue, for instance a perp exchange
- Protocols concentrating a large share of on-chain TVL
- Any method to quantify risk: graph analysis, simulation, statistical or agent-based

WHAT WE EXPECT

A working prototype that translates complex on-chain dependencies into a usable risk signal for portfolio management. Not a slide deck of an architecture, something that runs on real data and produces an output a manager could act on.

HOW WE WILL JUDGE IT

Criterion	Weight
Dependency discovery: depth and correctness	30%
Impact estimation: is the number defensible	30%
Decision usefulness: stay or exit, and why	25%
Working demo on real data	15%

WHAT BELEM BRINGS

Mentoring from our risk and on-chain engineering team throughout the event, office hours to challenge your modelling assumptions, and the portfolio structures we actually run as reference cases.

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